# Modeling the Regulatory Effects of Non-Coding Variants with AlphaGenome and CNN-Based Enhancer Classification

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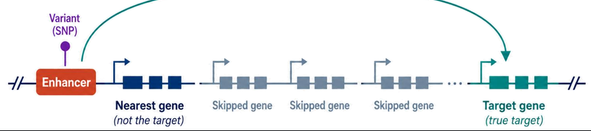
## Background

Non-coding variants can alter enhancer activity, but determining whether these regulatory changes affect downstream transcription, and which genes are involved, remains challenging.

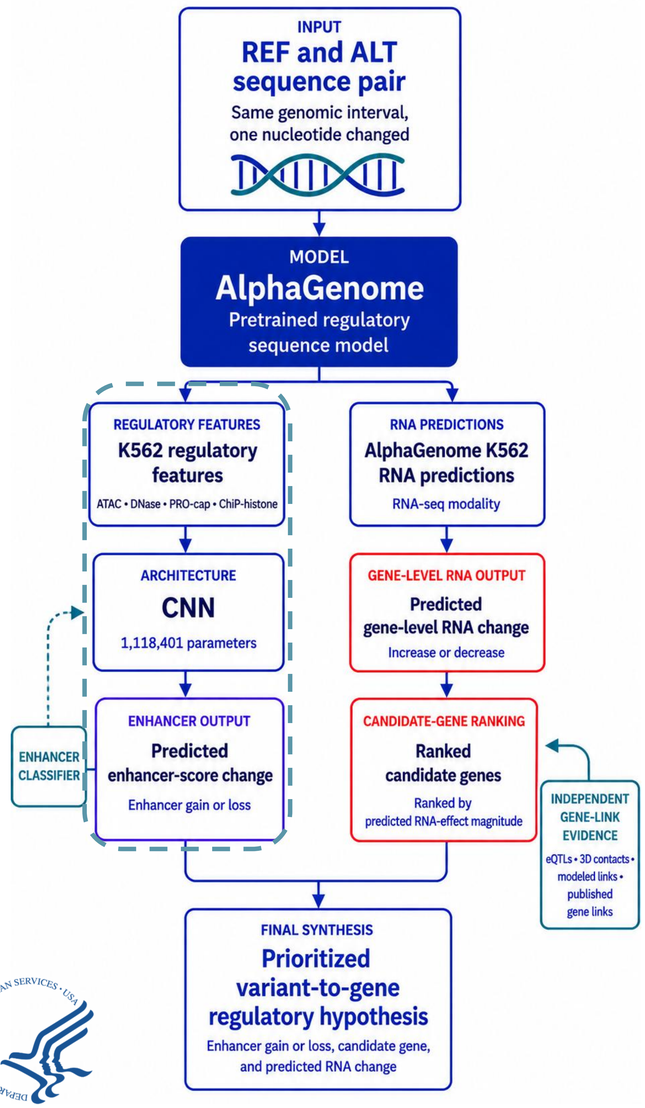



## Objective

Determine whether integrated K562 enhancer, AlphaGenome RNA, and independent gene-link evidence can prioritize target genes affected by non-coding variants.




## Framework




## K562 Enhancer Classifier and Variant Scoring

| 10,000 | 10,000 | ATAC<br>DNase<br>PRO-cap<br>ChIP-histone | 0.9803 | 0.9827 |
|---|---|---|---|---|
| K562 enhancer intervals | control intervals | | Validation AUROC | Validation AUPRC |

**29 threshold-crossing variants**




5% FPR threshold = 0.622

REF and ALT sequences were scored separately. Gain and loss were defined by crossing the 5% FPR enhancer threshold, while continuous ALT-minus-REF score changes were retained for directional analyses.

## Experimental Validation of Predicted Enhancer Effects

| 76% | 74% | 70% |
|---|---|---|
| K562 SuRE ↔ K562 Enhancer Classifier | K562 MPRA ↔ K562 Enhancer Classifier | All Three Same Direction |

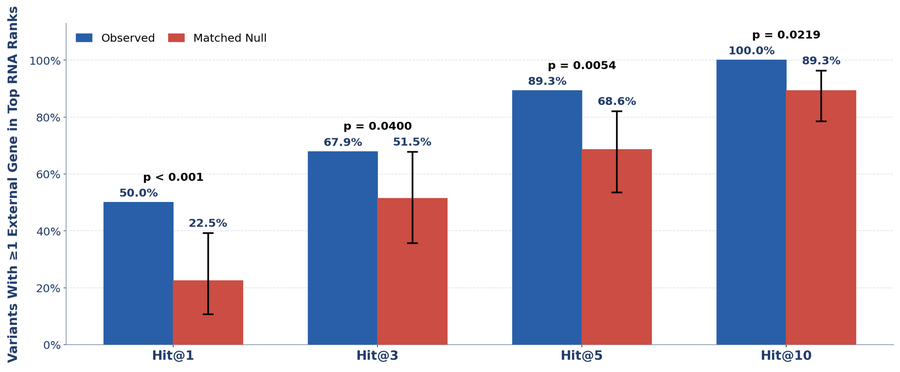
## Most Threshold-Crossing Variants Had Gene-Link Support

| 29/29<br>100.0%<br>Modeled gene-link evidence | 24/29<br>82.8%<br>Empirical gene-link evidence |
|---|---|
| 18/29<br>62.1%<br>Empirical + modeled same-gene support | 25/29<br>86.2%<br>External gene in RNA top five |

26/28
92.9%
Enhancer–RNA directional coherence

## Linking Enhancer Effects to Candidate Genes

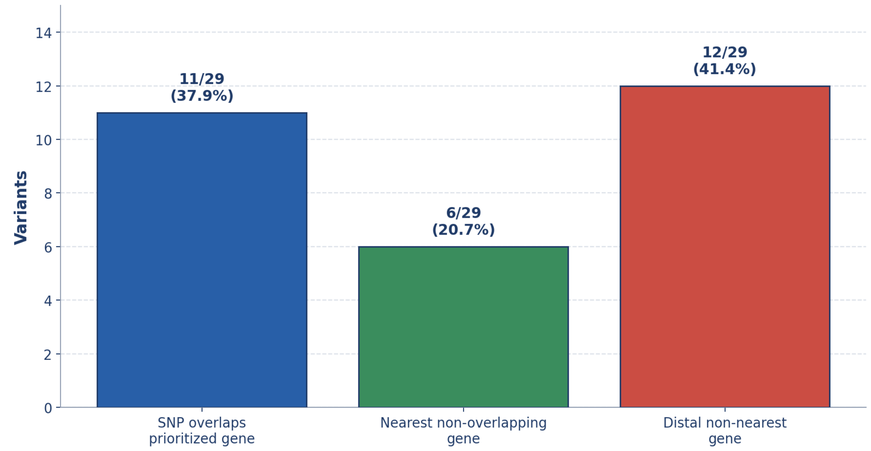
**Experimentally Supported Genes Rank Near the Top of AlphaGenome RNA Predictions**




For each SNP, genes independently nominated by eQTL, 3D-contact, or modeled enhancer–gene evidence were located within AlphaGenome's RNA-effect ranking and compared with a matched random baseline. Higher Hit@k values indicate that experimentally supported genes are captured near the top of AlphaGenome's predicted RNA-effect list.

## Integrated Target-Gene Prioritization

**Integrated Evidence Frequently Prioritized Distal Target Genes**




Integrated enhancer, RNA, and independent gene-link evidence frequently prioritized distal target genes, demonstrating that functional evidence extends beyond simple nearest-gene assignment.

## Representative Variant-to-Gene Hypotheses

| | | | |
|---|---|---|---|
| **Flagship gains**<br>**rs3748136 → PPP1R3B-DT** | **Enhancer gain:**<br>Δ = +0.911 | **Gene-link support:** eQTL, 3D, direct K562 editing | **RNA:** rank 1, predicted increase |
| **Strong convergent loss**<br>**rs1466990 → TEC** | **Enhancer loss:**<br>Δ = −0.975 | **Gene-link support:** eQTL, 3D, multiple modeled links | **RNA:** rank 1, predicted decrease |

## Conclusions

K562 enhancer classifier effects agreed with independent reporter-assay measurements.

Externally supported genes were enriched near the top of AlphaGenome RNA rankings. Integrated enhancer, gene-link, and RNA evidence generated prioritized variant-to-gene hypotheses.

Candidate-gene prioritization is evidence-guided rather than independent target validation.

## References

1. Avsec Ž et al. *Nature*. 2026;649:1206–1218. 2. van Arensbergen J et al. *Nat Genet*. 2019;51:1160–1169. 3. Siraj L et al. *Nature*. 2026. doi:10.1038/s41586-026-10121-6. 4. Moore JE et al. *Nature*. 2026. doi:10.1038/s41586-025-09909-9. 5. GTEx Consortium. *Science*. 2020;369:1318–1330.

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